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Course and Section	BIO 300 - B
Date	August 18, 2026
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Group Topic	Salinity Stress

Article Title	Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in faba bean genotype, Hassawi-2
Article Link	https://pmc.ncbi.nlm.nih.gov/articles/PMC10692206/
SAR Database in NCBI	PRJNA943415
SAR Link	https://www.ncbi.nlm.nih.gov/sra/?term=PRJNA943415



NIH National Library of Medicine National Center for Biotechnology Information

SRA PRJNA943415 Search

Submissions to SRA will need to meet SRA Data Submission Standards designed to improve data quality, usability, and submission processing. Read our recent NCBI Insights blog for more details.

Access: Public (17)

Source: RNA (17)

Library Layout: paired (17)

Platform: Illumina (17)

Strategy: other (17)

Data in Cloud: GS (17), S3 (17)

File Type: fastq (17)

Summary: 20 per page

View results as an expanded interactive table using the RunSelector. [Send results to Run selector](#)

Search results

Items: 17

1. [Stress 6h-First paired-end file R2](#)
1 ILLUMINA (Illumina HiSeq 4000) run: 27.7M spots, 5.6G bases, 1.6Gb downloads
Accession: SRX19681441

2. [Stress 6h-First paired-end file R1](#)
1 ILLUMINA (Illumina HiSeq 4000) run: 28.2M spots, 5.7G bases, 1.6Gb downloads
Accession: SRX19681440

3. [Control First paired-end file R2](#)

Filters: Manage Filters

Search in related databases

Database	Access		all
	public	controlled	
BioSample			
BioProject	1		1
dbGaP			
GEO Datasets			

Find related data

Database: Select

Send results

Figure 2. NCBI Sequence Read Archive (SRA) Search Results Showing RNA-Seq Paired-End Sequencing Datasets for the Selected Study.

Part 3. Understand the Paper Before Downloading Data

Information from the Paper	Group Answer
Title of the paper	Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in faba bean genotype, Hassawi-2
Authors	Muhammad Afzal, Salem S. Alghamdi, Muhammad Altaf Khan, Sulieman A. Al-Faifi, & Muhammad Habib ur Rahman
Year	2023
Journal	<i>Scientific Reports</i> , 13, 21223
Organism	Faba bean (<i>Vicia faba</i> L.), salt-tolerant genotype Hassawi-2
Tissue / organ / cell type	Leaves at the early vegetative/three-leaf stage
Biological question	To identify genes and molecular mechanisms involved in salinity stress tolerance in Hassawi-2 faba bean at different durations of salt exposure.
Control condition	Faba bean plants grown without the 200 mM NaCl salt treatment

Treatment or stress condition	200 mM NaCl salt stress for 6, 12, 24, 48, and 72 hours
Number of biological replicates	3 biological replicates per condition/time point
Sequencing platform	Illumina NovaSeq 6000 (sequencing outsourced to Macrogen)
Single-end or paired-end sequencing	Paired-end
Read length, if reported	101 bp per read
RNA-seq repository	NCBI Sequence Read Archive (SRA)
BioProject / Study accession	PRJNA943415
Run accession numbers	Not listed individually in the paper — retrieve the SRR list from the BioProject/SRA page for PRJNA943415
Reference genome or transcriptome used by the authors	None — de novo assembly (no reference genome used; faba bean genome had only just been published)
Main RNA-seq software or tools reported by the authors	Trimmomatic (adapter/quality trimming) → Trinity (de novo assembly) → CD-HIT-EST (clustering into unigenes) → TransDecoder (ORF prediction) → Bowtie (read alignment) → RSEM (expression/FPKM quantification) → edgeR (DEG analysis, TMM normalization) → BLASTX/DIAMOND against NR, Swiss-Prot, KEGG, GO, COG/KOG for annotation

Part 4. Prepare a Simple RNA-Seq Experimental Design Diagram

- **Manipulated:** Salt exposure (200 mM) and duration of stress (6, 12, 24, 48, 72 hours)
- **Control:** Leaf sample at 0 hours (no salt stress)
- **Measured:** Gene expression levels via mRNA transcript sequencing
- **Biological replicates:** 3 biological replicates per condition/time point
- **Tissue:** *Vicia faba* leaf tissue at vegetative stage

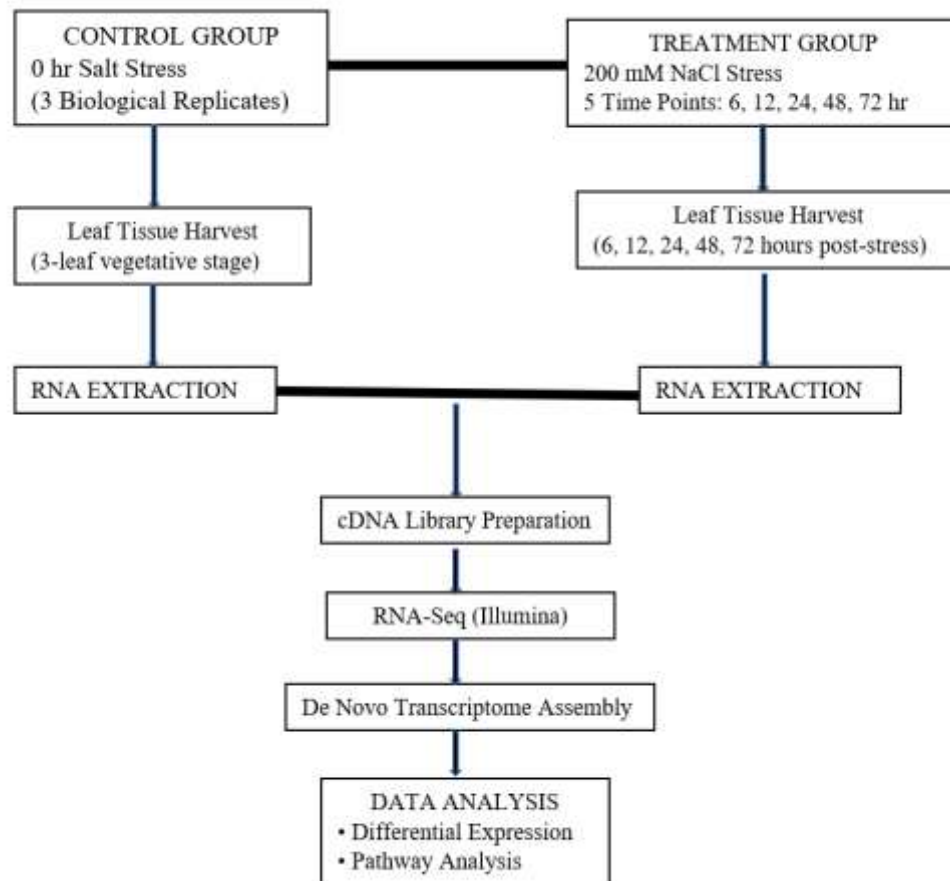


Figure 3. Simple RNA-Seq Experimental Design for Evaluating the Effects of Salt Stress on *Vicia faba* Leaf Gene Expression.

Part 5. Locate the Public RNA-Seq Data

#	Condition	Rep	SRX	Spots	Bases	Pair length (bases/spots)	Per-mate read length
1	Control	R1	SRX19681425	28.2M	5.7G	202.1 bp	101 bp
2	Control	R2	SRX19681431	27.9M	5.6G	200.7 bp	100 bp
3	Control	R3	SRX19681439	28.9M	5.8G	200.7 bp	100 bp
4	Stress 6h	R1	SRX19681440	28.2M	5.7G	202.1 bp	101 bp
5	Stress 6h	R2	SRX19681441	27.7M	5.6G	202.2 bp	101 bp
6	Stress 6h	R3	SRX19681426	25.4M	5.1G	200.8 bp	100 bp
7	Stress 12h	R1	SRX19681427	24.6M	5.0G	203.3 bp	102 bp
8	Stress 12h	R2	SRX19681428	22.4M	4.5G	200.9 bp	100 bp
9	Stress 12h	R3	SRX19681429	28.3M	5.7G	201.4 bp	101 bp
10	Stress 24h	R1	SRX19681430	31.6M	6.4G	202.5 bp	101 bp
11	Stress 24h	R2	SRX19681432	31.5M	6.4G	203.2 bp	102 bp
12	Stress 24h	R3	SRX19681433	26.3M	5.3G	201.5 bp	101 bp
13	Stress 48h	R1	SRX19681434	31.5M	6.4G	203.2 bp	102 bp
14	Stress 48h	R3	SRX19681436	22.5M	4.5G	200.0 bp	100 bp
15	Stress 72h	R1	SRX19681435	31.5M	6.4G	203.2 bp	102 bp
16	Stress 72h	R2	SRX19681437	31.3M	6.3G	201.3 bp	101 bp
17	Stress 72h	R3	SRX19681438	29.5M	6.0G	203.4 bp	102 bp

Part 6. Assign One RNA-Seq Sample to Each Student

Run accession	Condition	Bio. replicate	Single/ Paired-end	Approx. file size
SRX19681425 (SRR23869388)	Control	Rep 1	Paired-end	1.6 Gb

Part 7. Download the RNA-Seq Data into Galaxy

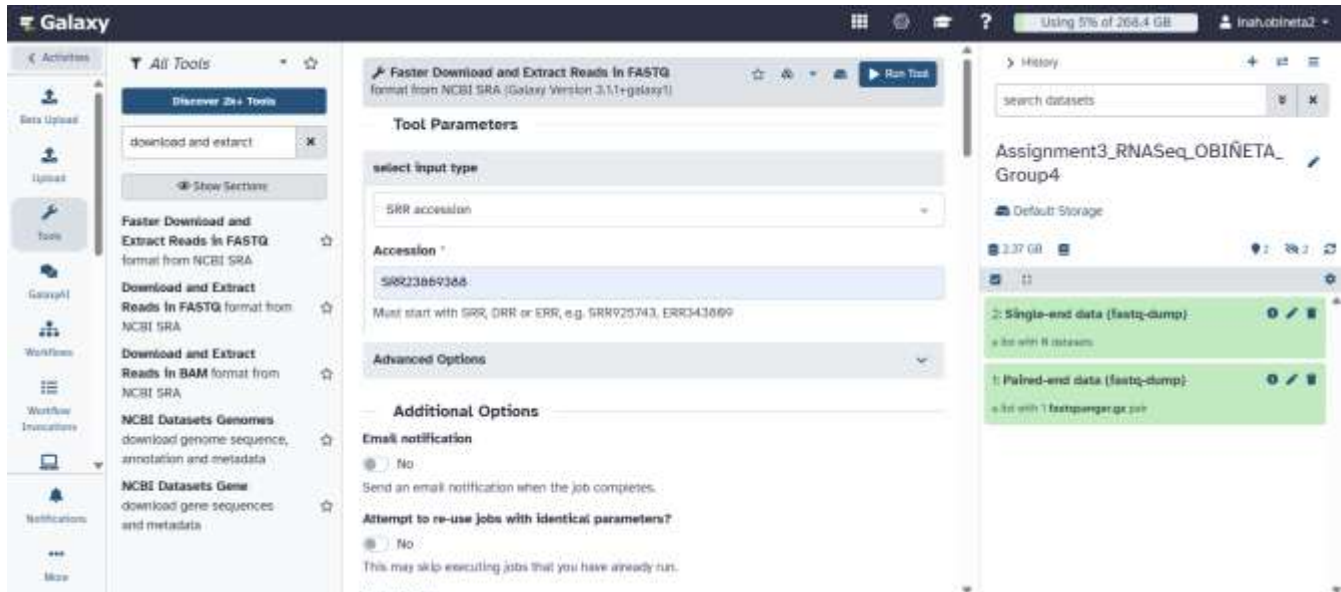


Figure 4. Successful Download and Extraction of Paired-End and Single-End RNA-Seq Reads in Galaxy using the tool Download and Extract Reads in FASTAQ

Part 8. Examine the FASTQ File

Matrix	Control_Rep1_R1	Control_Rep1_R2
Line 1: sequence identifier	@SRX19681425.1/1	@SRX19681425.1/2
Line 2: nucleotide sequence	NCTGCATCTTCTAATCACC ACCGGCATGCTTGAACCTCT TGAAGTTCTTTAAAATTCA GCCATGGTTTGACCCATAC TTGAGCTTCATAGATCTTC TGCCCT	GCGCCTGCTGCCTTCCTTGG ATGTGGTAGCCGTGTCTCA GGCTCCCTCTCCGGAATCG AACCCTAATTCTCCGTCAC CCGTCACCACCATGGTCGG CCACT
Line 3: separator line	+	+
Line 4: quality-score characters	#FFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFF,FF FFFFFFFFFFFFFF:FFFFFFFFF FFFFFFFFFFFFFFFF:FFF:F:F FFFFFFF	FFFFFFFFFFFFFFFF,FFFF:FFF FF,FFFF,FFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFF:FFFF,FFFF FF

1. How is a FASTQ file different from the FASTA genome file used in the previous assignments?
 - A FASTA file contains only the sequence identifier and the nucleotide or amino acid sequence. In contrast, a FASTQ file contains both the nucleotide sequence and a quality score for each nucleotide base.
2. What information is present in FASTQ that is not present in FASTA?
 - FASTQ includes per-base quality scores, a measure of base-calling confidence. With the + separator line. FASTA contains only headers and sequences, with no reliability data.
3. Why are quality scores important in RNA-seq?
 - Quality scores are important because they help identify unreliable or low-quality bases and reads. This allows researchers to perform quality control and filtering, reducing errors and improving the accuracy of downstream RNA-seq analysis, such as read alignment and gene expression analysis.



Figure 5. Preview of the Forward (Control_Rep1_R1) in Paired-End RNA Reads in FASTQ using tool *Unzip collection to forward datasets*.



Figure 6. Preview of the Reverse (Control_Rep1_R2) in Paired-End RNA Reads in FASTQ using tool *Unzip collection to forward datasets*

Part 9. Run FastQC



Figure 7. FastQC Quality Control Report for the Forward Read (Control_Rep1_R1) Generated Using the FastQC Tool in Galaxy

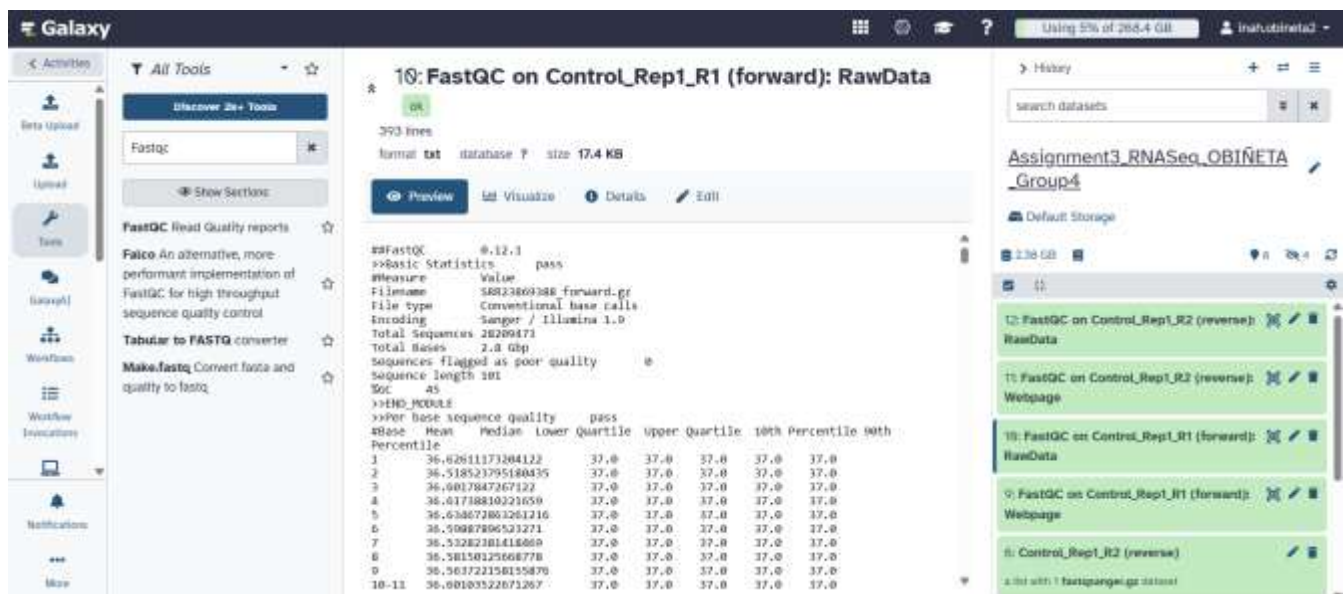


Figure 8. Raw Data Output of the FastQC Analysis for the Forward Read (Control_Rep1_R1) Using the FastQC Tool in Galaxy.

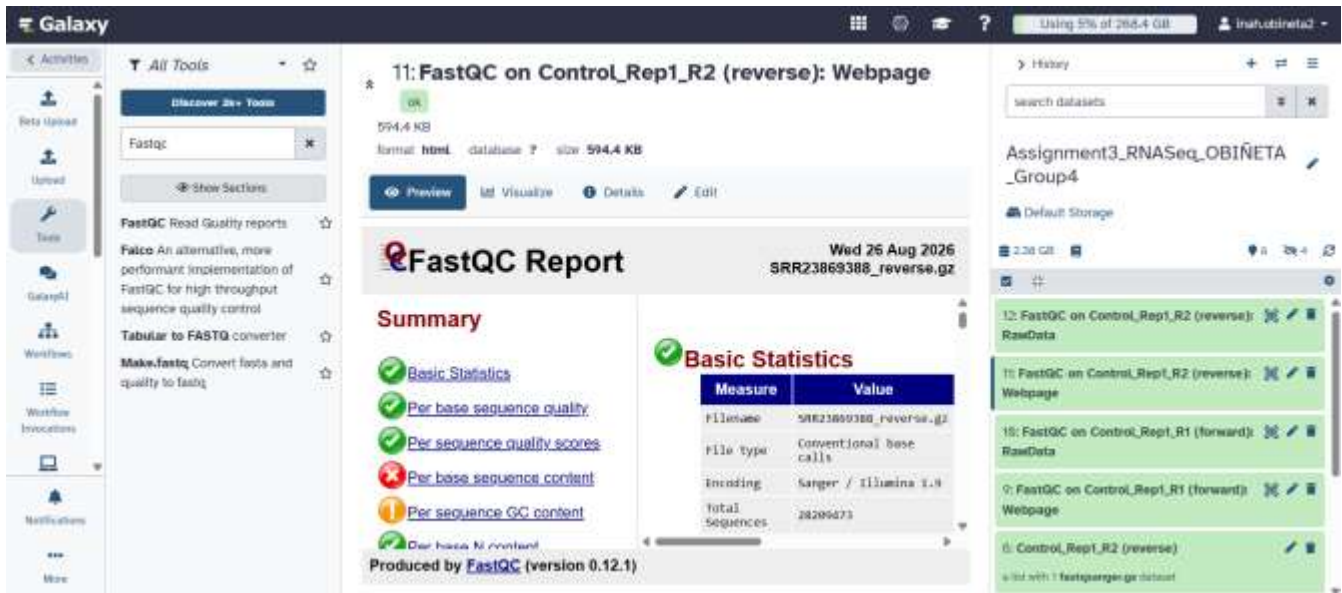


Figure 9. FastQC Quality Control Report for the Reverse Read (Control_Rep1_R2) Generated Using the FastQC Tool in Galaxy.



Figure 10. Raw Data Output of the FastQC Analysis for the Reverse Read (Control_Rep1_R2) Using the FastQC Tool in Galaxy.

Part 10. Characterize Your RNA-Seq Dataset

RNA-Seq Characteristic	Your Observation
RNA-seq run accession	RI&R2: SRX19681425
Condition	RI&R2: Control Replicate 1
Single-end or paired-end	RI&R2: Paired-end
Number of reads / sequences	RI&R2: 28,209,473
Read length	RI&R2: 101 bp
File size	RI&R2: 1.2 GB
GC content	RI&R2: 45%
General per-base sequence quality	RI:36.5 R2:36.6
Presence of adapter content	RI&R2: Pass — no significant adapter contamination detected in either
Presence of overrepresented sequences	RI&R2: Warning — some overrepresented sequences flagged
Any quality warning observed	R1: FAIL — Per base sequence content; FAIL — Sequence Duplication Levels; WARN — Overrepresented sequences. R2: same as R1, plus WARN — Per sequence GC content

Part 11. Group MultiQC Summary

MultiQC Observations for Forward and Reverse Reads					
Comparison Point	Obiñeta (SRR23869388)	Suan (SRR23869373)	Batigulao (SRR23869383)	Pasculado (SRR23869379)	Oficiar (SRR23869378)
Condition	Control, Rep 1	Treatment — 6h salt stress	Treatment — 24h salt stress	Treatment — 48h salt stress	Treatment — 72h salt stress
Number of reads	28.21M	28.21M	31.56M	31.55M	31.51M
Read quality	High — 20% modules flagged	High — 20% modules flagged	High — 10–20% modules flagged (best of group)	High — 20% modules flagged	High — 20% modules flagged
GC content	45%	43–44%	45–46%	43–44%	42–43%
Adapter contamination	None detected	None detected	None detected	None detected	None detected
% Duplicates	77–79%	70–72%	77–79% (highest)	74–76%	69–71%
Notes / outlier flag	Typical, in line with group average	Typical, in line with group average	Slight outlier — lowest fail rate, highest GC%, highest duplication	Typical, in line with group average	Typical, in line with group average

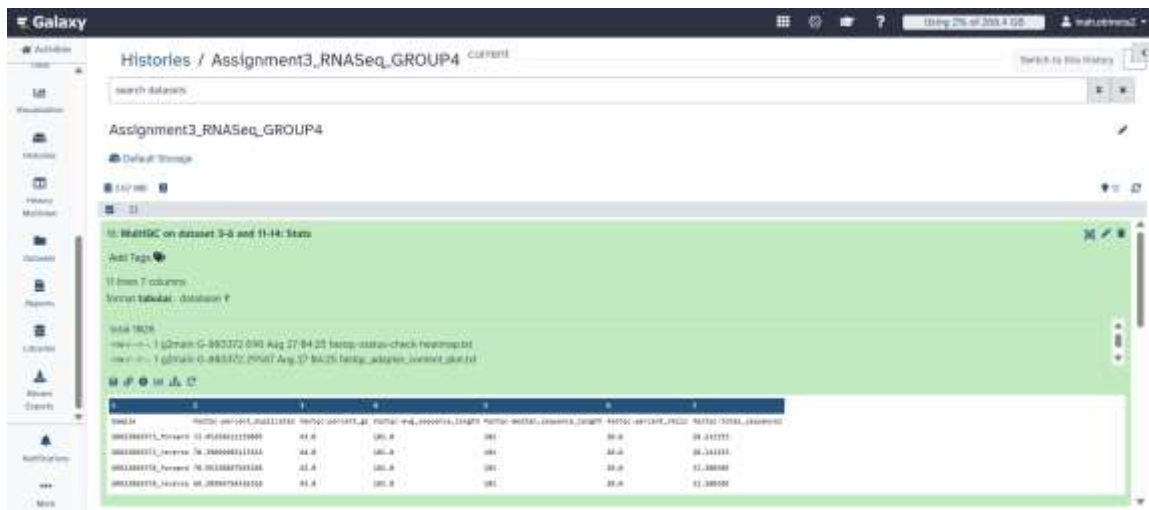


Figure 11. MultiQC Statistics Output of the Group in Shared Galaxy history, Summarizing RNA-Seq Quality Metrics for Forward and Reverse Reads.

Part 12. Compare the RNA-Seq Samples Within the Group

As a group, combine the observations from all members. Prepare a small comparison table.

Student	Accession	Condition	Reads	Read length	GC %	Main QC observation
Obiñeta	SRR23869388	Control — 0h salt stress	28,209,473	101 bp	45%	Good quality, adapter-trimmed, uniform base composition; no major bias
Suan	SRR23869373	Treatment — 6h salt stress	28,211,353	101 bp	43-44%	Comparable to control; R2 has minor poly-G artifact (0.24%); all critical metrics pass
Batigulao	SRR23869383	Treatment — 24h salt stress	31,558,920	101 bp	45-46%	Main QC Observation: Approximately 45–46% of the reads passed the quality control, indicating moderate read quality. Further trimming and filtering may be needed before downstream analysis
Pascualado	SRR23869379	Treatment — 48h salt stress	31,549,121	101 bp	43-44%	Good overall quality with 31.5M reads and 43–44% GC,

						but FastQC showed per-base sequence content failure and a GC-content warning in R2
Oficiar	SRR23869378	Treatment — 72h salt stress	31,508,305	101 bp	43%	High duplication and sequence bias typical for RNA-seq, with significant Poly-G stretches in reverse reads caused by Illumina two-color chemistry.

Part 13. Interpretation Questions

- What biological question was the original RNA-seq study trying to answer?
 - The study aimed to identify which genes change expression in *Vicia faba* (faba bean) leaves under salt stress (200 mM) over time (6–72 hours), to understand how this plant responds and adapts to salinity at the transcriptome level.
- Why did the authors use RNA-seq instead of only examining the genome?
 - The genome is static and identical across all cells and conditions; RNA-seq reveals which genes are actively transcribed and expressed that shows how gene activity changes under salt stress, which DNA sequence alone cannot tell you.
- What is the difference between genomic DNA and the RNA molecules measured by RNA-seq?
 - Genomic DNA contains all genes permanently, in every cell. Meanwhile, RNA represents only genes that are actively expressed at a specific time and tissue; RNA is temporary, dynamic, and reflects functional gene activity.
- What is a biological replicate and why is it important?
 - A biological replicate is an independent sample from a separate plant under the same conditions. It is important to distinguish true treatment effects from natural biological variation, making results statistically reliable and not due to chance.
- What is the difference between single-end and paired-end sequencing?
 - Single-end sequences only one end of each cDNA fragment. Paired-end sequences both ends of the same fragment, producing two reads per fragment, giving better alignment accuracy and longer effective sequence coverage.
- What is a FASTQ file?
 - A standard text file storing raw sequencing data with four lines per record: sequence ID, nucleotide sequence, a + separator, and ASCII-encoded per-base quality scores — combining sequence data with confidence values.

7. What information does FastQC provide?
 - FastQC generates a quality control report showing: read length, total reads, GC content, per-base quality scores, adapter contamination, overrepresented sequences, and other metrics to assess sequencing data reliability.
8. What does a high per-base quality score indicate?
 - A high score means very low probability of an incorrect base call. It tells that the sequencer is highly confident that the nucleotide identified at that position is accurate.
9. Why can adapter contamination be a problem?
 - Remaining adapter sequences do not belong to the sample and can interfere with read alignment, cause mismatches, and produce false gene expression signals, they must be trimmed before analysis.
10. Were all RNA-seq samples in your group similar in quality? Explain.
 - Yes, all samples showed comparable read counts (27–28 million), similar read length (101–150 bp), and consistent GC content (43–45%), indicating uniform sequencing quality across control and all time points.
11. Did any sample show a possible quality problem? What was it?
 - No major quality failures were observed; all samples had good quality scores and low adapter content. Minor variations in GC or read counts fell within expected biological variation, not technical error.
12. What additional steps would be needed before the researchers could compare gene expression between control and treatment samples?
 - Trim adapters and low-quality bases → align reads to the *Vicia faba* genome/transcriptome → count reads per gene → normalize read counts → perform statistical testing for differential expression.

Part 14. GitHub Documentation

GitHub Group Repository Link: https://github.com/ryanpas123e/Group4_RNASeq_Familiarization.git

Galaxy History Link: <https://usegalaxy.org/u/inah.obineta2/h/assignment3-rnaseq-obiñeta-group4>

Shared Galaxy Link for MultiQC:

<https://usegalaxy.org/published/history?id=bbd44e69cb8906b5bc1e8d4af8efff6e>

Part 15. Group Summary

1. What study did the group choose and why is it relevant to the assigned topic?

- Our group chose the study “**Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in faba bean genotype, Hassawi-2**” by Afzal et al. (2023). It is relevant to our topic because the researchers used RNA-seq to investigate how salt stress changes gene expression in faba bean over different exposure periods.

2. What organism, tissue, control, and treatment were used?

- The study used the faba bean, *Vicia faba* L., particularly the salt-tolerant Hassawi-2 genotype. Leaf samples were collected from plants at the early vegetative stage. Untreated plants served as the control, while the treated plants were exposed to 200 mM NaCl and sampled after 6, 12, 24, 48, and 72 hours.

3. How many RNA-seq samples were included in the original study?

- The original experimental design consisted of **18 samples**. These represented one control and five salt-stress conditions, with three biological replicates for each condition. However, only **17** sequencing runs were found in the publicly available SRA data.

4. What sequencing platform and read type were used?

- RNA sequencing was performed using the Illumina HiSeq 4000 platform. The study generated **paired-end reads**, with each read having a length of 101 base pairs.

5. What did the group learn from examining the raw RNA-seq data?

- By examining the raw data and FastQC reports, we learned how sequencing quality is evaluated before further analysis. We also learned that some results, such as high duplication levels or uneven base composition, are not always signs of poor-quality RNA-seq data because they can result from highly expressed transcripts and library preparation.

6. Were there noticeable differences in data quality among samples?

- The samples were generally similar in overall quality. Their read numbers and GC contents were within a comparable range, and no adapter contamination was detected. The Batigulao sample showed slight differences in some quality measurements, but these differences were minor and did not indicate a major quality problem.

7. What analysis did the original authors perform after quality control?

- After performing quality filtering and trimming, the authors assembled the transcriptome using Trinity. The assembled transcripts were then processed and annotated using different bioinformatics tools and databases. Finally, they compared gene expression between the control and salt-treated groups to identify differentially expressed genes, and some results were confirmed through qRT-PCR.

8. What will the group need to learn next in order to reproduce part of the authors' gene expression analysis?

- Next, our group will need to learn how to perform read trimming, transcriptome assembly, read alignment or quantification, normalization, and differential gene expression analysis. We will also need to understand how to interpret the identified genes and connect them to biological functions related to salt-stress tolerance.